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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 26th June Evening Shift

JEE Main Question Paper with Answer Key

26 Jun 2022	90	18	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	22 questions
Physics	26 questions
Out of syllabus	18 questions
Answer key	at the end

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Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

The number of radial and angular nodes in 4d orbital are, respectively

(A) 1 and 2 (B) 3 and 2 (C) 1 and 0 (D) 2 and 1

Open solution

Q2. MCQ (Single Correct Answer)

Match List-I with List-II

	List - I Enzyme		List - II Conversion of
A.	Invertase	I.	Starch into maltose
B.	Zymase	II.	Maltose into glucose
C.	Diastase	III.	Glucose into ethanol
D.	Maltase	IV.	Cane sugar into glucose

Choose the most appropriate answer from the options given below :

(A) A-III, B-IV, C-II, D-I (B) A-III, B-II, C-I, D-IV

(C) A-IV, B-III, C-I, D-II (D) A-IV, B-II, C-III, D-I

Open solution

Q3. MCQ (Single Correct Answer)

Which of the following elements is considered as a metalloid?

(A) Sc (B) Pb (C) Bi (D) Te

Open solution

Q4. MCQ (Single Correct Answer)

The oxide which contains an odd electron at the nitrogen atom is :

(A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5

Open solution

Q5. MCQ (Single Correct Answer)

Which one of the following is an example of disproportionation reaction ?

(A) $3MnO_4^{2-} + 4H^+ \rightarrow 2MnO_4^- + MnO_2 + 2H_2O$

(B) $MnO_4^- + 4H^+ + 4e^- \rightarrow MnO_2 + 2H_2O$

(C) $10I^- + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5I_2$

(D) $8MnO_4^- + 3S_2O_3^{2-} + H_2O \rightarrow 8MnO_2 + 6SO_4^{2-} + 2OH^-$

Open solution

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Q6. MCQ (Single Correct Answer)

The most common oxidation state of Lanthanoid elements is +3. Which of the following is likely to deviate easily from +3 oxidation state?

(A) Ce (At. No. 58) (B) La (At. No. 57) (C) Lu (At. No. 71) (D) Gd (At. No. 64)

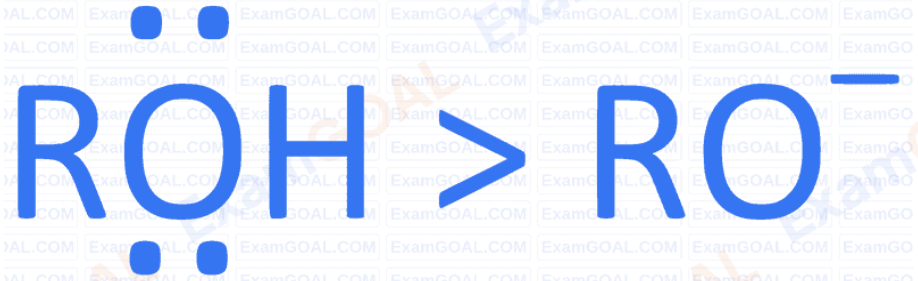
Open solution

Q7. MCQ (Single Correct Answer)

The correct order of nucleophilicity is

(A) $F^- > OH^-$

(B) 

(C) 

(D) $NH_2^- > NH_3$

Open solution

Q8. MCQ (Single Correct Answer)

Oxidation of toluene to benzaldehyde can be easily carried out with which of the following reagents?

(A) CrO_3/acetic acid, H_3O^+ (B) CrO_3/acetic anhydride, H_3O^+

(C) $KMnO_4$/HCl, H_3O^+ (D) CO/HCl, anhydrous $AlCl_3$

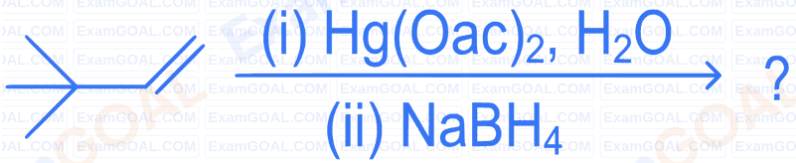
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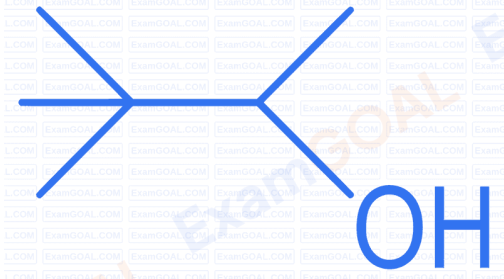
Q9. MCQ (Single Correct Answer)

Open solution

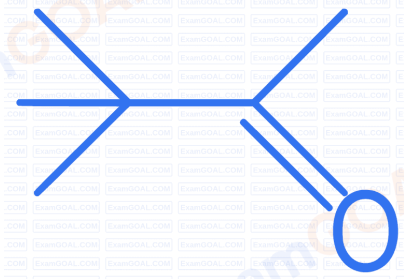
The major product in the following reaction



(A)



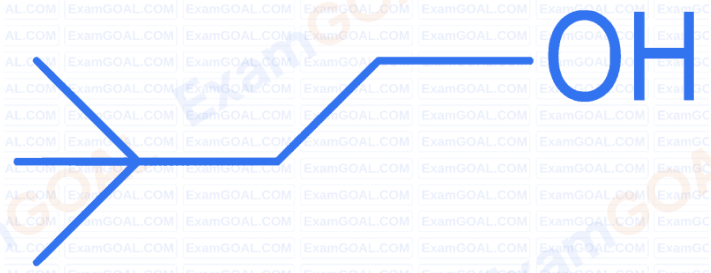
(B)



(C)



(D)



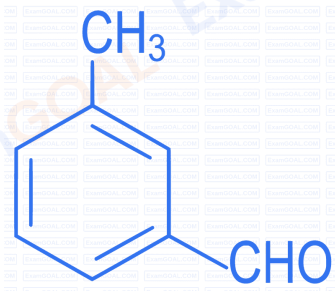
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Q10. MCQ (Single Correct Answer)

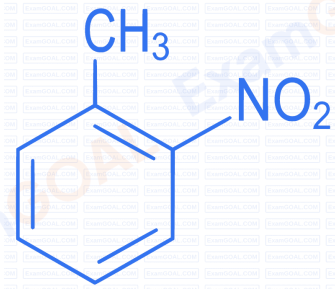
Open solution

Halogenation of which one of the following will yield m-substituted product with respect to methyl group as a major product?

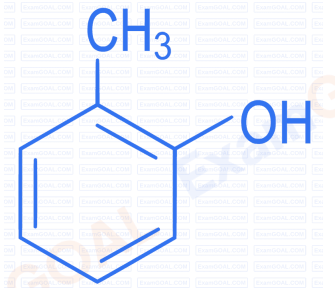
(A)



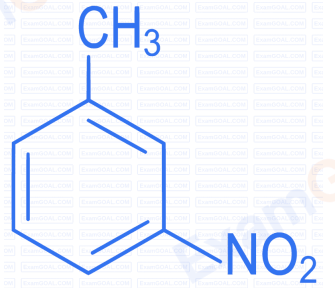
(B)



(C)



(D)



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Q11. MCQ (Single Correct Answer)

Open solution

The reagent, from the following, which converts benzoic acid to benzaldehyde in one step is



(A) $LiAlH_4$ (B) $KMnO_4$ (C) MnO (D) $NaBH_4$

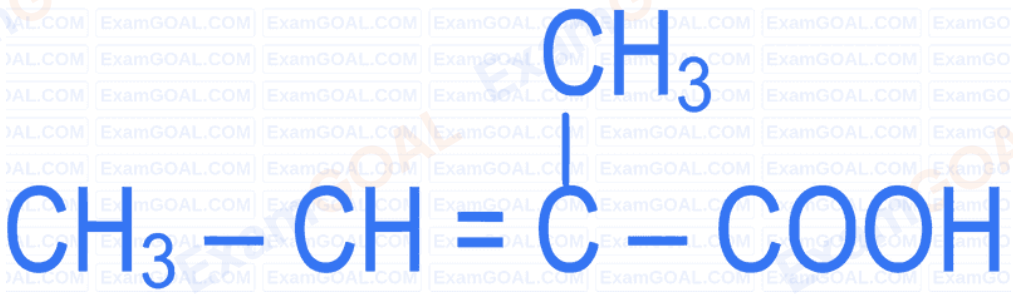
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Q12. MCQ (Single Correct Answer)

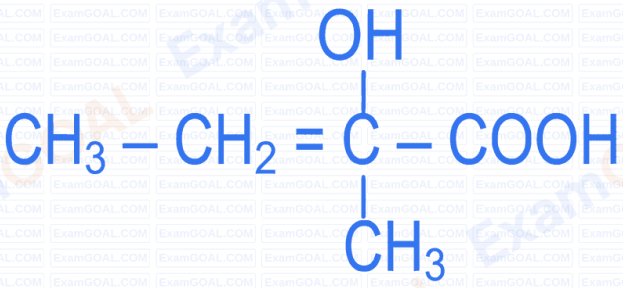
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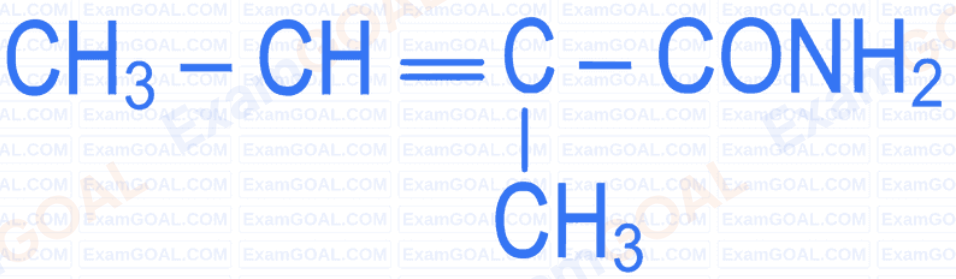
The final product 'A' in the following reaction sequence



(A) 

(B) 

(C) 

(D) 

Q13. MCQ (Single Correct Answer)

Open solution

Which statement is NOT correct for p-toluenesulphonyl chloride?

(A) It is known as Hinsberg's reagent.

(B) It is used to distinguish primary and secondary amines.

(C) On treatment with secondary amine, it leads to a product, that is soluble in alkali.

(D) It doesn't react with tertiary amines.

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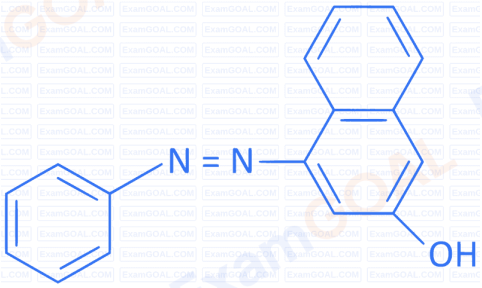
Q14. MCQ (Single Correct Answer)

Open solution

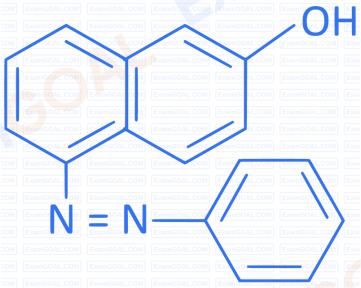
The final product 'C' in the following series of reactions



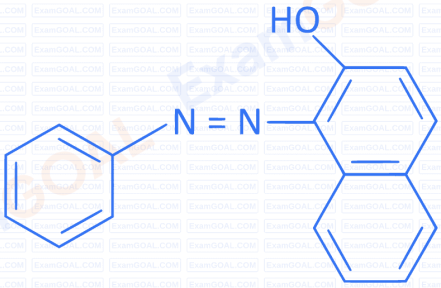
(A)



(B)



(C)



(D)



Q15. MCQ (Single Correct Answer)

Open solution

Which one of the following is a water soluble vitamin, that is not excreted easily?

(A) Vitamin B_2 (B) Vitamin B_1 (C) Vitamin B_6 (D) Vitamin B_{12}

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Q16. Numerical

CNG is an important transportation fuel. When 100 g CNG is mixed with 208 g oxygen in vehicles, it leads to the formation of CO_2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is _____. [nearest integer]

[Assume CNG to be methane]

Open solution

Q17. Numerical

Amongst SF_4, XeF_4, CF_4 and H_2O, the number of species with two lone pairs of electrons is _____.

Open solution

Q18. Numerical

A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36 g. When it is subjected to cooking at 100°C, then the internal energy for vaporization in kJ mol^{-1} is _____. [nearest integer]

[Assume steam to be an ideal gas. Given $\Delta_{vap}H^{\Theta}$ for water at 373 K and 1 bar is 41.1 kJ mol^{-1} ; R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q19. Numerical

The osmotic pressure exerted by a solution prepared by dissolving 2.0 g of protein of molar mass 60 kg mol^{-1} in 200 mL of water at 27°C is _____ Pa. [integer value]

(use R = 0.083 L bar mol^{-1} K^{-1})

Open solution

Q20. Numerical

40% of HI undergoes decomposition to H_2 and I_2 at 300 K. ΔG^{Θ} for this decomposition reaction at one atmosphere pressure is _____ J mol^{-1}. [nearest integer]

(Use R = 8.31 J K^{-1} mol^{-1} ; log 2 = 0.3010, ln 10 = 2.3, log 3 = 0.477)

Open solution

Q21. Numerical

$Cu(s) + Sn^{2+}$ (0.001M) $\rightarrow$ Cu^{2+} (0.01M) + Sn(s)

The Gibbs free energy change for the above reaction at 298 K is x $\times$ 10^{-1} kJ mol^{-1}. The value of x is _____. [nearest integer]

[Given : $E^{\Theta}_{Cu^{2+}/Cu} = 0.34\,V$; $E^{\Theta}_{Sn^{2+}/Sn} = -0.14\,V$; F = 96500 C mol^{-1}]

Open solution

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Q22. Numerical

Catalyst A reduces the activation energy for a reaction by 10 kJ mol^{-1} at 300 K. The ratio of rate constants, $\frac{k_{T,\,Catalysed}}{k_{T,\,Uncatalysed}}$ is e^x. The value of x is _____. [nearest integer]

[Assume that the pre-exponential factor is same in both the cases. Given R = 8.31 J K^{-1} mol^{-1}]

Open solution

Q23. Numerical

Reaction of $[Co(H_2O)_6]^{2+}$ with excess ammonia and in the presence of oxygen results into a diamagnetic product. Number of electrons present in t_{2g}-orbitals of the product is _____.

Open solution

Q24. Numerical

The moles of methane required to produce 81 g of water after complete combustion is _____ $\times 10^{-2}$ mol. [nearest integer]

Open solution

Mathematics 22 questions

Q25. MCQ (Single Correct Answer)

Let f : R $\rightarrow$ R be defined as f (x) = x $-$ 1 and g : R $-$ {1, $-$1} $\rightarrow$ R be defined as $g(x) = \frac{x^2}{x^2-1}$.

Then the function fog is :

(A) one-one but not onto

(B) onto but not one-one

(C) both one-one and onto

(D) neither one-one nor onto

Open solution

Q26. MCQ (Single Correct Answer)

If the system of equations

αx + y + z = 5, x + 2y + 3z = 4, x + 3y + 5z = β

has infinitely many solutions, then the ordered pair (α, β) is equal to :

(A) (1, $-$3)

(B) ($-$1, 3)

(C) (1, 3)

(D) ($-$1, $-$3)

Open solution

Q27. MCQ (Single Correct Answer)

$\lim\limits_{x\to 0} \frac{\cos(\sin x)-\cos x}{x^4}$ is equal to :

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

(C) $\frac{1}{6}$

(D) $\frac{1}{12}$

Open solution

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Q28. MCQ (Single Correct Answer)

Let f(x) = min {1, 1 + x sin x}, $0 \leq x \leq 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to

(A) (2, 0) (B) (1, 0) (C) (1, 1) (D) (2, 1)

Open solution

Q29. MCQ (Single Correct Answer)

Consider a cuboid of sides 2x, 4x and 5x and a closed hemisphere of radius r. If the sum of their surface areas is a constant k, then the ratio x : r, for which the sum of their volumes is maximum, is :

(A) 2 : 5 (B) 19 : 45 (C) 3 : 8 (D) 19 : 15

Open solution

Q30. MCQ (Single Correct Answer)

The area of the region bounded by $y^2 = 8x$ and $y^2 = 16(3 - x)$ is equal to:

(A) $\frac{32}{3}$ (B) $\frac{40}{3}$ (C) 16 (D) 19

Open solution

Q31. MCQ (Single Correct Answer)

If $\int \frac{1}{x}\sqrt{\frac{1-x}{1+x}}dx = g(x) + c, g(1) = 0$, then $g\left(\frac{1}{2}\right)$ is equal to :

(A) $\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) + \frac{\pi}{3}$

(B) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) + \frac{\pi}{3}$

(C) $\log_e\left(\frac{\sqrt{3}+1}{\sqrt{3}-1}\right) - \frac{\pi}{3}$

(D) $\frac{1}{2}\log_e\left(\frac{\sqrt{3}-1}{\sqrt{3}+1}\right) - \frac{\pi}{6}$

Open solution

Q32. MCQ (Single Correct Answer)

If $y = y(x)$ is the solution of the differential equation

$x\frac{dy}{dx} + 2y = x\,e^x, y(1) = 0$ then the local maximum value

of the function $z(x) = x^2y(x) - e^x$, $x \in R$ is :

(A) 1 − e (B) 0 (C) $\frac{1}{2}$ (D) $\frac{4}{e} - e$

Open solution

Q33. MCQ (Single Correct Answer)

If the solution of the differential equation

$\frac{dy}{dx} + e^x\left(x^2 - 2\right)y = \left(x^2 - 2x\right)\left(x^2 - 2\right)e^{2x}$ satisfies $y(0) = 0$, then the value of y(2) is _____.

(A) -1 (B) 1 (C) 0 (D) e

Open solution

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Q34. MCQ (Single Correct Answer)

The locus of the mid point of the line segment joining the point (4, 3) and the points on the ellipse $x^2 + 2y^2 = 4$ is an ellipse with eccentricity :

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{1}{2\sqrt{2}}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{2}$

Open solution

Q35. MCQ (Single Correct Answer)

Let $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}, \vec{b} = 2\hat{i} - 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$ be three given vectors. Let $\vec{v}$ be a vector in the plane of $\vec{a}$ and $\vec{b}$ whose projection on $\vec{c}$ is $\frac{2}{\sqrt{3}}$. If $\vec{v}.\ \hat{j} = 7$, then $\vec{v}.\ \left(\hat{i} + \hat{k}\right)$ is equal to :

(A) 6 (B) 7 (C) 8 (D) 9

Open solution

Q36. MCQ (Single Correct Answer)

The mean and standard deviation of 50 observations are 15 and 2 respectively. It was found that one incorrect observation was taken such that the sum of correct and incorrect observations is 70. If the correct mean is 16, then the correct variance is equal to :

(A) 10 (B) 36 (C) 43 (D) 60

Open solution

Q37. MCQ (Single Correct Answer)

$16\sin(20^\circ)\sin(40^\circ)\sin(80^\circ)$ is equal to :

(A) $\sqrt{3}$ (B) $2\sqrt{3}$ (C) 3 (D) $4\sqrt{3}$

Open solution

Q38. MCQ (Single Correct Answer)

If the inverse trigonometric functions take principal values then

$\cos^{-1}\left(\frac{3}{10}\cos\left(\tan^{-1}\left(\frac{4}{3}\right)\right) + \frac{2}{5}\sin\left(\tan^{-1}\left(\frac{4}{3}\right)\right)\right)$ is equal to :

(A) 0 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

Open solution

Q39. Numerical

Let f : R $\rightarrow$ R satisfy $f(x + y) = 2^x f(y) + 4^y f(x)$, $\forall$x, y $\in$ R. If f(2) = 3, then 14. $\frac{f'(4)}{f'(2)}$ is equal to _____.

Open solution

Q40. Numerical

Let p and q be two real numbers such that p + q = 3 and $p^4 + q^4 = 369$. Then $\left(\frac{1}{p} + \frac{1}{q}\right)^{-2}$ is equal to _____.

Open solution

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Q41. Numerical

If $z^2 + z + 1 = 0, z \in C$, then

$\left|\sum_{n=1}^{15}\left(z^n + (-1)^n \frac{1}{z^n}\right)^2\right|$ is equal to _____.

Open solution

Q42. Numerical

Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $Y = \alpha I + \beta X + \gamma X^2$ and $Z = \alpha^2 I - \alpha\beta X + (\beta^2 - \alpha\gamma)X^2$, $\alpha, \beta, \gamma \in$ R. If $Y^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$, then $(\alpha - \beta + \gamma)^2$ is equal to _____.

Open solution

Q43. Numerical

The total number of 3-digit numbers, whose greatest common divisor with 36 is 2, is _____.

Open solution

Q44. Numerical

If a_1 (> 0), a_2, a_3, a_4, a_5 are in a G.P., $a_2 + a_4 = 2a_3 + 1$ and $3a_2 + a_3 = 2a_4$, then $a_2 + a_4 + 2a_5$ is equal to _____.

Open solution

Q45. Numerical

The integral $\frac{24}{\pi}\int_0^{\sqrt{2}} \frac{(2-x^2)dx}{(2+x^2)\sqrt{4+x^4}}$ is equal to _____.

Open solution

Q46. Numerical

If the probability that a randomly chosen 6-digit number formed by using digits 1 and 8 only is a multiple of 21 is p, then 96 p is equal to _____.

Open solution

Physics 26 questions

Q47. MCQ (Single Correct Answer)

The dimension of mutual inductance is :

(A) $[ML^2T^{-2}A^{-1}]$ (B) $[ML^2T^{-3}A^{-1}]$ (C) $[ML^2T^{-2}A^{-2}]$ (D) $[ML^2T^{-3}A^{-2}]$

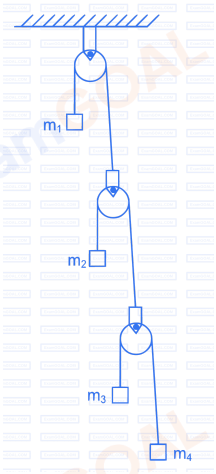
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Q48. MCQ (Single Correct Answer)

Open solution

In the arrangement shown in figure a_1, a_2, a_3 and a_4 are the accelerations of masses m_1, m_2, m_3 and m_4 respectively. Which of the following relation is true for this arrangement?



(A) $4a_1 + 2a_2 + a_3 + a_4 = 0$

(B) $a_1 + 4a_2 + 3a_3 + a_4 = 0$

(C) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

(D) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

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Q49. MCQ (Single Correct Answer)

Arrange the four graphs in descending order of total work done; where W_1, W_2, W_3 and W_4 are the work done corresponding to figure a, b, c and d respectively.

Open solution

Figure - a

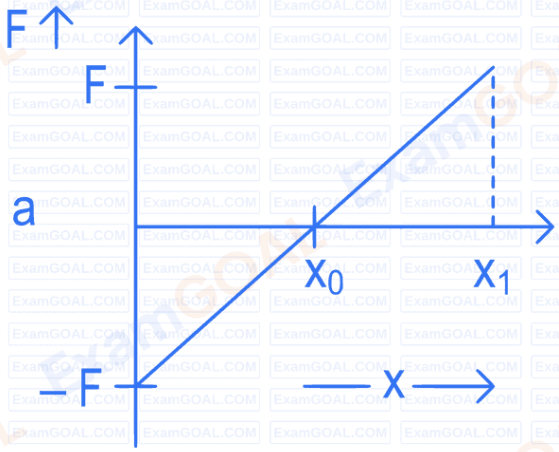


Figure - b

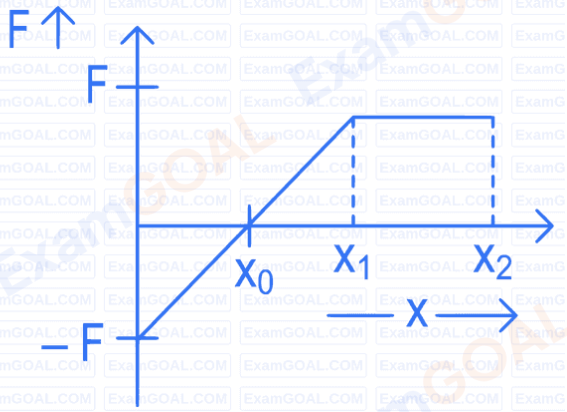
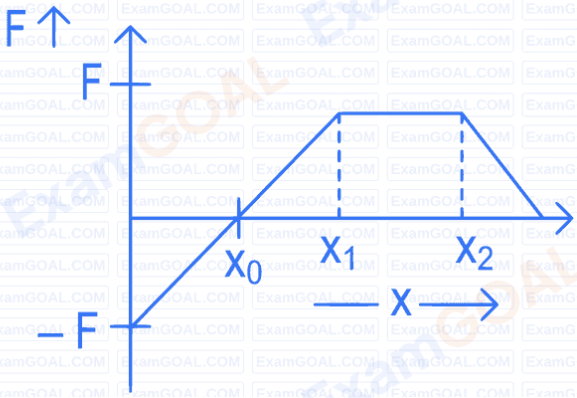
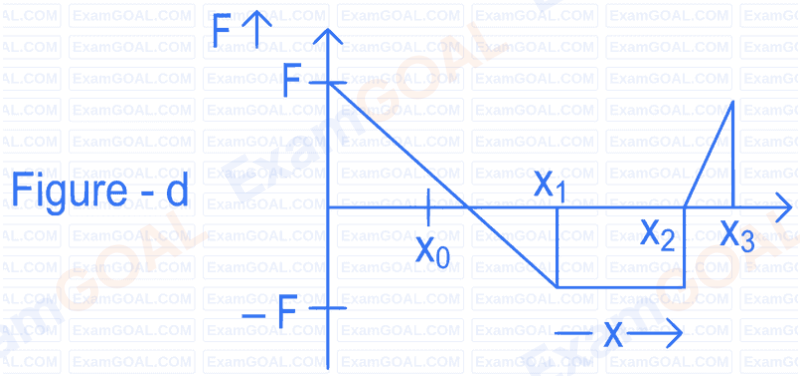


Figure - c



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(A) $W_3 > W_2 > W_1 > W_4$

(B) $W_3 > W_2 > W_4 > W_1$

(C) $W_2 > W_3 > W_4 > W_1$

(D) $W_2 > W_3 > W_1 > W_4$

Q50. MCQ (Single Correct Answer)

A solid spherical ball is rolling on a frictionless horizontal plane surface about its axis of symmetry. The ratio of rotational kinetic energy of the ball to its total kinetic energy is

(A) $\frac{2}{5}$ (B) $\frac{2}{7}$ (C) $\frac{1}{5}$ (D) $\frac{7}{10}$

Open solution

Q51. MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.

Reason R : At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below:

(A) Both A and R are true and R is the correct explanation of A.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) A is true but R is false.

(D) A is false but R is true.

Open solution

Q52. MCQ (Single Correct Answer)

If ρ is the density and η is coefficient of viscosity of fluid which flows with a speed v in the pipe of diameter d, the correct formula for Reynolds number R_e is :

(A) $R_e = \frac{\eta d}{\rho v}$ (B) $R_e = \frac{\rho v}{\eta d}$

(C) $R_e = \frac{\rho v d}{\eta}$ (D) $R_e = \frac{\eta}{\rho v d}$

Open solution

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Q53. MCQ (Single Correct Answer)

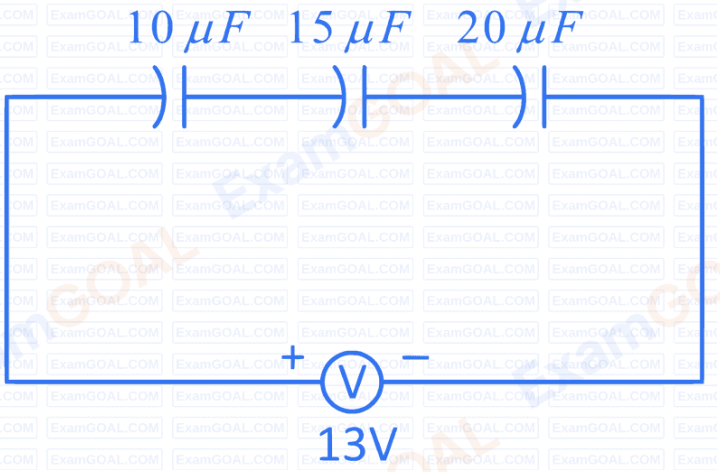
A flask contains argon and oxygen in the ratio of 3 : 2 in mass and the mixture is kept at 27°C. The ratio of their average kinetic energy per molecule respectively will be :

(A) 3 : 2 (B) 9 : 4 (C) 2 : 3 (D) 1 : 1

Open solution

Q54. MCQ (Single Correct Answer)

The charge on capacitor of capacitance 15μF in the figure given below is :

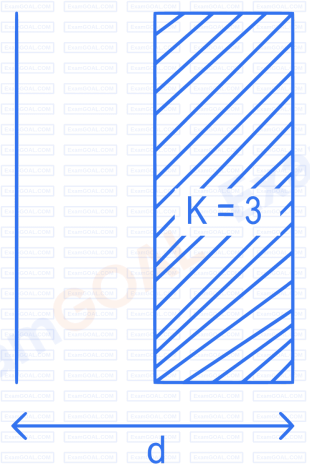


(A) 60μC (B) 130μC (C) 260μC (D) 585μC

Open solution

Q55. MCQ (Single Correct Answer)

A parallel plate capacitor with plate area A and plate separation d = 2 m has a capacitance of 4 μF. The new capacitance of the system if half of the space between them is filled with a dielectric material of dielectric constant K = 3 (as shown in figure) will be :



(A) 2μF (B) 32μF (C) 6μF (D) 8μF

Open solution

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Q56. MCQ (Single Correct Answer)

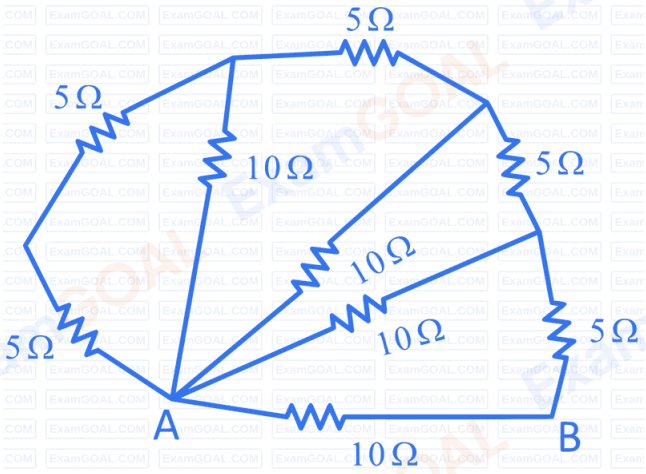
Sixty four conducting drops each of radius 0.02 m and each carrying a charge of 5 μC are combined to form a bigger drop. The ratio of surface charge density of bigger drop to the smaller drop will be :

(A) 1 : 4 (B) 4 : 1 (C) 1 : 8 (D) 8 : 1

Open solution

Q57. MCQ (Single Correct Answer)

The equivalent resistance between points A and B in the given network is :



(A) 65Ω (B) 20Ω (C) 5Ω (D) 2Ω

Open solution

Q58. MCQ (Single Correct Answer)

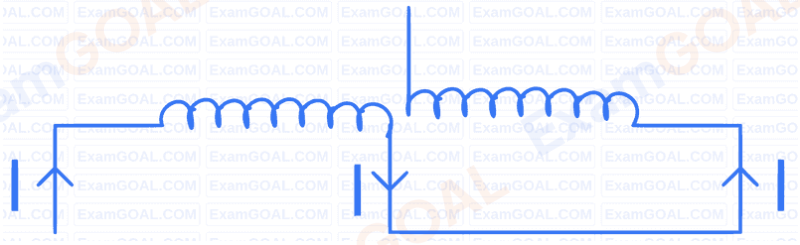
A bar magnet having a magnetic moment of 2.0×10^5 JT^{-1}, is placed along the direction of uniform magnetic field of magnitude B = 14×10^{-5} T. The work done in rotating the magnet slowly through 60° from the direction of field is :

(A) 14 J (B) 8.4 J (C) 4 J (D) 1.4 J

Open solution

Q59. MCQ (Single Correct Answer)

Two coils of self inductance L_1 and L_2 are connected in series combination having mutual inductance of the coils as M. The equivalent self inductance of the combination will be :



(A) $\frac{1}{L_1}+\frac{1}{L_2}+\frac{1}{M}$ (B) L_1+L_2+M

(C) L_1+L_2+2M (D) L_1+L_2-2M

Open solution

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Q60. MCQ (Single Correct Answer)

A metallic conductor of length 1 m rotates in a vertical plane parallel to east-west direction about one of its end with angular velocity 5 rad s^{-1}. If the horizontal component of earth's magnetic field is 0.2×10^{-4} T, then emf induced between the two ends of the conductor is :

(A) 5 μV (B) 50 μV (C) 5 mV (D) 50 mv

Open solution

Q61. MCQ (Single Correct Answer)

Which is the correct ascending order of wavelengths?

(A) $\lambda_{visible} < \lambda_{X\text{-}ray} < \lambda_{gamma\text{-}ray} < \lambda_{microwave}$ (B) $\lambda_{gamma\text{-}ray} < \lambda_{X\text{-}ray} < \lambda_{visible} < \lambda_{microwave}$

(C) $\lambda_{X\text{-}ray} < \lambda_{gamma\text{-}ray} < \lambda_{visible} < \lambda_{microwave}$ (D) $\lambda_{microwave} < \lambda_{visible} < \lambda_{gamma\text{-}ray} < \lambda_{X\text{-}ray}$

Open solution

Q62. MCQ (Single Correct Answer)

For a specific wavelength 670 nm of light coming from a galaxy moving with velocity v, the observed wavelength is 670.7 nm. The value of v is :

(A) 3×10^{8} ms^{-1} (B) 3×10^{10} ms^{-1}

(C) 3.13×10^{5} ms^{-1} (D) 4.48×10^{5} ms^{-1}

Open solution

Q63. MCQ (Single Correct Answer)

A metal surface is illuminated by a radiation of wavelength 4500 $\overset{o}{A}$. The ejected photo-electron enters a constant magnetic field of 2 mT making an angle of 90° with the magnetic field. If it starts revolving in a circular path of radius 2 mm, the work function of the metal is approximately :

(A) 1.36 eV (B) 1.69 eV (C) 2.78 eV (D) 2.23 eV

Open solution

Q64. Numerical

A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at t = 0s. At t = 2s, another ball is projected vertically upward with same velocity. At t = _____ s, second ball will meet the first ball (g = 10 ms^{-2}).

Open solution

Q65. Numerical

A batsman hits back a ball of mass 0.4 kg straight in the direction of the bowler without changing its initial speed of 15 ms^{-1}. The impulse imparted to the ball is _____ Ns.

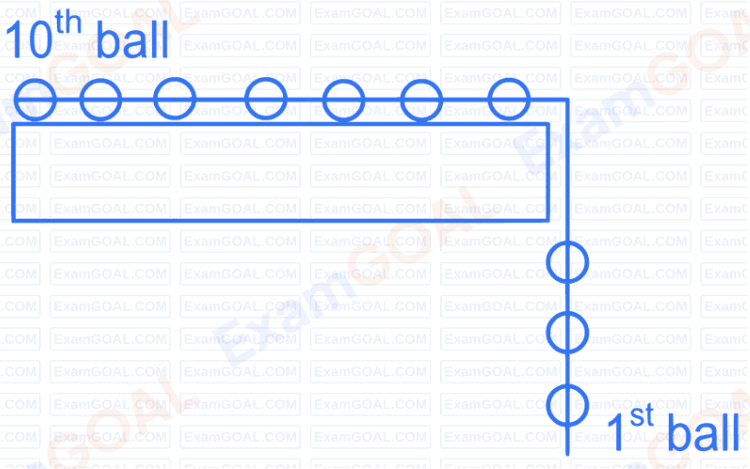
Open solution

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Q66. Numerical

A system of 10 balls each of mass 2 kg are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the 7th and 8th ball is _____ N when 6th ball just leaves the table.

Open solution



Q67. Numerical

A geyser heats water flowing at a rate of 2.0 kg per minute from 30°C to 70°C. If geyser operates on a gas burner, the rate of combustion of fuel will be _____ g min^{-1}.

[Heat of combustion = 8×10^3 Jg^{-1}, Specific heat of water = 4.2 Jg^{-1} $°C^{-1}$]

Open solution

Q68. Numerical

A set of 20 tuning forks is arranged in a series of increasing frequencies. If each fork gives 4 beats with respect to the preceding fork and the frequency of the last fork is twice the frequency of the first, then the frequency of last fork is _____ Hz.

Open solution

Q69. Numerical

Two 10 cm long, straight wires, each carrying a current of 5A are kept parallel to each other. If each wire experienced a force of 10^{-5} N, then separation between the wires is _____ cm.

Open solution

Q70. Numerical

A small bulb is placed at the bottom of a tank containing water to a depth of $\sqrt{7}$ m. The refractive index of water is $\frac{4}{3}$. The area of the surface of water through which light from the bulb can emerge out is $x\pi$ m^2. The value of x is _____.

Open solution

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Q71. Numerical

A travelling microscope is used to determine the refractive index of a glass slab. If 40 divisions are there in 1 cm on main scale and 50 Vernier scale divisions are equal to 49 main scale divisions, then least count of the travelling microscope is _____ $\times 10^{-6}$ m.

Open solution

Q72. Numerical

The stopping potential for photoelectrons emitted from a surface illuminated by light of wavelength 6630 $\overset{o}{A}$ is 0.42 V. If the threshold frequency is x $\times 10^{13}$ /s, where x is _____ (nearest integer).

(Given, speed light = 3 $\times 10^{8}$ m/s, Planck's constant = 6.63 $\times 10^{-34}$ Js)

Open solution

Out of syllabus 18 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q73. MCQ (Single Correct Answer) · **Chemistry**

The role of depressants in 'Froth Flotation method' is to

(A) selectively prevent one component of the ore from coming to the froth.

(B) reduce the consumption of oil for froth formation.

(C) stabilize the froth.

(D) enhance non-wettability of the mineral particles.

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Boiling of hard water is helpful in removing the temporary hardness by converting calcium hydrogen carbonate and magnesium hydrogen carbonate to :

(A) $CaCO_3$ and $Mg(OH)_2$

(B) $CaCO_3$ and $MgCO_3$

(C) $Ca(OH)_2$ and $MgCO_3$

(D) $Ca(OH)_2$ and $Mg(OH)_2$

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

s-block element which cannot be qualitatively confirmed by the flame test is :

(A) Li (B) Na (C) Rb (D) Be

Open solution

Q76. MCQ (Single Correct Answer) · **Chemistry**

The measured BOD values for four different water samples (A-D) are as follows: A = 3 ppm; B = 18 ppm; C = 21 ppm; D = 4 ppm. The water samples which can be called as highly polluted with organic wastes, are :

(A) A and B (B) A and D (C) B and C (D) B and D

Open solution

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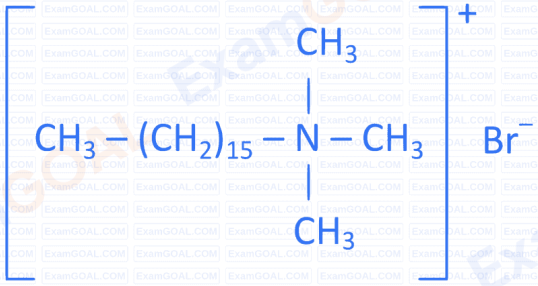
Q77. MCQ (Single Correct Answer) · **Chemistry**

Which of the following is NOT an example of synthetic detergent?

Open solution

(A) 

(B) $CH_3-(CH_2)_{16}-COO^-Na^+$

(C) 

(D) $CH_3(CH_2)_{16}COO(CH_2CH_2O)_nCH_2CH_2OH$

Q78. Numerical · **Chemistry**

In a solid AB, A atoms are in ccp arrangement and B atoms occupy all the octahedral sites. If two atoms from the opposite faces are removed, then the resultant stoichiometry of the compound is A_xB_y. The value of x is _____. [nearest integer]

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

If $A=\sum_{n=1}^{\infty}\frac{1}{(3+(-1)^n)^n}$ and $B=\sum_{n=1}^{\infty}\frac{(-1)^n}{(3+(-1)^n)^n}$, then $\frac{A}{B}$ is equal to :

(A) $\frac{11}{9}$ (B) 1 (C) $-\frac{11}{9}$ (D) $-\frac{11}{3}$

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

If m is the slope of a common tangent to the curves $\frac{x^2}{16}+\frac{y^2}{9}=1$ and $x^2+y^2=12$, then $12m^2$ is equal to :

(A) 6 (B) 9 (C) 10 (D) 12

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

The normal to the hyperbola

$\frac{x^2}{a^2}-\frac{y^2}{9}=1$ at the point $\left(8,3\sqrt{3}\right)$ on it passes through the point :

(A) $\left(15,-2\sqrt{3}\right)$ (B) $\left(9,2\sqrt{3}\right)$ (C) $\left(-1,9\sqrt{3}\right)$ (D) $\left(-1,6\sqrt{3}\right)$

Open solution

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Q82. MCQ (Single Correct Answer) · **Mathematics**

If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point :

(A) (2, −2, 0) (B) (−2, 2, 0) (C) (1, 0, 2) (D) (−1, 0, −2)

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If the lines $\vec{r} = \left(\hat{i} - \hat{j} + \hat{k}\right) + \lambda\left(3\hat{j} - \hat{k}\right)$ and $\vec{r} = \left(\alpha\hat{i} - \hat{j}\right) + \mu\left(2\hat{i} - 3\hat{k}\right)$ are co-planar, then the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is :

(A) $\frac{2}{9}$ (B) $\frac{2}{11}$ (C) $\frac{4}{11}$ (D) 2

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let r $\in$ {p, q, $\sim$p, $\sim$q} be such that the logical statement

r $\vee$ ($\sim$p) $\Rightarrow$ (p $\wedge$ q) $\vee$ r

is a tautology. Then r is equal to :

(A) p (B) q (C) $\sim$p (D) $\sim$q

Open solution

Q85. Numerical · **Mathematics**

If $\left({}^{40}C_0\right) + \left({}^{41}C_1\right) + \left({}^{42}C_2\right) + \ldots\ldots + \left({}^{60}C_{20}\right) = \frac{m}{n}{}^{60}C_{20}$ m and n are coprime, then m + n is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

Let a line L_1 be tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{4} = 1$ and let L_2 be the line passing through the origin and perpendicular to L_1. If the locus of the point of intersection of L_1 and L_2 is $(x^2 + y^2)^2 = \alpha x^2 + \beta y^2$, then $\alpha + \beta$ is equal to _____.

Open solution

Q87. MCQ (Single Correct Answer) · **Physics**

A radioactive nucleus can decay by two different processes. Half-life for the first process is 3.0 hours while it is 4.5 hours for the second process. The effective half-life of the nucleus will be:

(A) 3.75 hours (B) 0.56 hours (C) 0.26 hours (D) 1.80 hours

Open solution

Q88. MCQ (Single Correct Answer) · **Physics**

The positive feedback is required by an amplifier to act an oscillator. The feedback here means :

(A) External input is necessary to sustain ac signal in output.

(B) A portion of the output power is returned back to the input.

(C) Feedback can be achieved by LR network.

(D) The base-collector junction must be forward biased.

Open solution

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Q89. MCQ (Single Correct Answer) · **Physics**

A sinusoidal wave y(t) = 40sin(10 × 10^6 πt) is amplitude modulated by another sinusoidal wave x(t) = 20sin (1000 πt). The amplitude of minimum frequency component of modulated signal is :

(A) 0.5 (B) 0.25 (C) 20 (D) 10

Open solution

Q90. Numerical · **Physics**

A heat engine operates with the cold reservoir at temperature 324 K. The minimum temperature of the hot reservoir, if the heat engine takes 300 J heat from the hot reservoir and delivers 180 J heat to the cold reservoir per cycle, is _____ K.

Open solution

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Answer Key

Chemistry

1. **A**	2. **C**	3. **D**	4. **B**	5. **A**
6. **A**	7. **D**	8. **B**	9. **A**	10. **C**
11. **C**	12. **A**	13. **C**	14. **C**	15. **D**
16. **143**	17. **1**	18. **38**	19. **415**	20. **2735**
21. **983**	22. **4**	23. **6**	24. **225**	

Mathematics

25. **D**	26. **C**	27. **C**	28. **B**	29. **B**
30. **C**	31. **A**	32. **D**	33. **C**	34. **C**
35. **D**	36. **C**	37. **B**	38. **C**	39. **248**
40. **4**	41. **2**	42. **100**	43. **150**	44. **40**
45. **3**	46. **33**			

Physics

47. **C**	48. **A**	49. **A**	50. **B**	51. **D**
52. **C**	53. **D**	54. **A**	55. **C**	56. **B**
57. **C**	58. **A**	59. **D**	60. **B**	61. **B**
62. **C**	63. **A**	64. **6**	65. **12**	66. **36**
67. **42**	68. **152**	69. **5**	70. **9**	71. **5**
72. **35**				

Out of syllabus

73. **A**	74. **A**	75. **D**	76. **C**	77. **B**
78. **3**	79. **C**	80. **B**	81. **C**	82. **C**
83. **B**	84. **C**	85. **102**	86. **12**	87. **D**
88. **B**	89. **D**	90. **540**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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